

ENTHUSIAST ADVANCE COURSE

PHASE : MEB-I (A)

TARGET : PRE MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 14-08-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	1	3	3	2	3	2	4	4	3	2	3	1	2	1	3	3	2	2	2	2	3	4	1	4	3	1	3	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	3	3	1	3	3	4	3	2	2	3	3	2	3	1	1	1	1	4	4	3	2	4	2	4	3	4	3	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	3	4	2	2	1	2	1	2	1	1	4	1	1	2	1	4	2	2	2	1	4	3	1	2	3	1	3	1	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	3	3	1	3	1	1	4	2	3	2	3	3	3	4	2	2	4	4	3	1	4	2	3	2	2	3	4	1	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	2	1	3	2	3	3	3	4	4	1	1	2	2	4	2	1	1	2	3	3	4	4	1	1	4	3	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	2	1	1	2	3	2	2	3	4	2	3	3	3	1	4	3	4	1	2	3	2	2	3	3	2	3	1	4	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	2	2	4	4	2	2	1	3	1	4	2	4	3	3	2	4	2	3	3										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

2. Ans (1)

$$\beta = \frac{0.4\text{mm}}{4/3} = 0.3 \text{ mm}$$

3. Ans (3)

$$\omega_x = \frac{2\lambda D}{a}$$

4. Ans (3)

$$I_{\text{output}} = \frac{I_0}{2} \cos^2 60 = \frac{I_0}{8}$$

= 12.5% of I_0

5. Ans (2)

$$\sin \theta = \frac{\Delta x}{a}$$

for 1st minimum $\rightarrow$

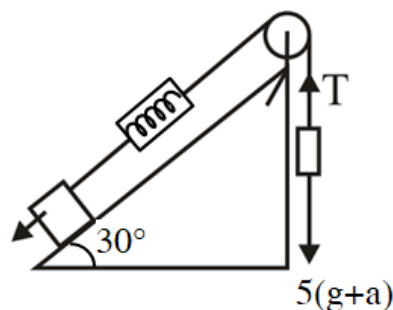
$$\Delta x = \lambda$$

$$\sin 30^\circ = \frac{\lambda}{a}$$

$$\frac{1}{2} = \frac{\lambda}{a}$$

$$\lambda = \frac{5}{2} = 2.5 \text{ cm}$$

6. Ans (3)

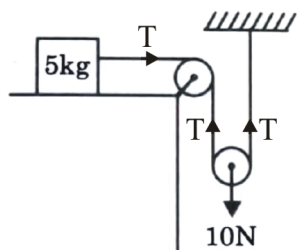


Both side forces are same so block is in equilibrium.

$$T = 5 (g + a)$$

$$= 5(12) = 60 \text{ N}$$

7. **Ans (2)**



At pulley

$$2T = 10$$

$$T = 5 \text{ N}$$

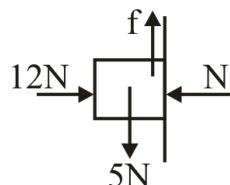
For block

$$T = 5a$$

$$8 = 5a$$

$$a = 1 \text{ m/s}^2$$

8. **Ans (4)**



$$f_L = \mu_S N$$

$$= 0.5 \times 12$$

$$= 6 \text{ N}$$

$$\because Mg < f_L$$

$\therefore$ Block is rest

$$f_S = 5 \text{ N}$$

$$N = 12 \text{ N}$$

$$\text{Contact force} = \sqrt{F^2 + N^2} = 13 \text{ N}$$

9. **Ans (4)**

$$t = \frac{\mu}{\mu g} \Rightarrow 10 = \frac{6}{\mu(10)}$$

$$\mu = 0.06$$

10. **Ans (3)**

By work energy theorem $w_{\text{gravity}} + w_{\text{friction}} = \Delta \text{KE}$

$$mgR + w_{\text{friction}} = \Delta \text{KE}$$

$$mgR + w_{\text{friction}} = \frac{1}{2} mV^2 - 0$$

$$w_{\text{friction}} = \frac{1}{2} (2) (8)^2 - (2) (10) (5)$$

$$= 64 - 100$$

$$= -36 \text{ J}$$

11. **Ans (2)**

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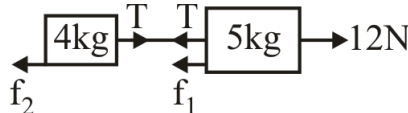
If we consider two blocks together then total

$$\text{friction} = m(m_1 + m_2)g$$

$$= 0.2(9) (10)$$

$$= 18 \text{ N}$$

Means system cannot move, so



For 5 kg block

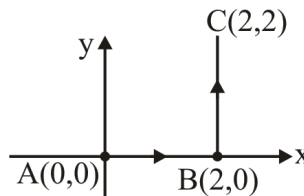
$$T + f_1 = 12 \text{ N}$$

$$T = 12 - \mu(5g)$$

$$T = 12 - 10$$

$$T = 2 \text{ N}$$

12. **Ans (3)**



For A to B

$$w_1 = \int_0^2 2x dx = [x^2]_0^2$$

$$= 4 \text{ J}$$

For B to C

$$w_2 = \int_0^2 3y dy = \left[\frac{3y^2}{2} \right]_0^2$$

$$= 6 \text{ J}$$

$$w = 4 + 6 = 10 \text{ J}$$

13. **Ans (1)**

$$\frac{du}{dx} = 6x - 6x^2 = 0 \rightarrow x = 0$$

$$6x(1-x) = 0 \rightarrow x = 1$$

$$\text{Now } \frac{d^2u}{dx^2} = 6 - 12x$$

$$\text{At } x = 0, \frac{d^2u}{dx^2} \Rightarrow \oplus \text{ means stable}$$

$$x = 1, \frac{d^2u}{dx^2} \Rightarrow \ominus \text{ means unstable}$$

14. **Ans (2)**

$$w = \left(\frac{m}{4} \right) (g) \left(\frac{L}{8} \right)$$

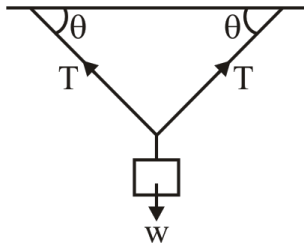
$$w = \frac{MgL}{32}$$

15. Ans (1)

$$KE = \frac{p^2}{2M} \text{ As } p = \text{same}$$

$$KE \propto \frac{1}{M}$$

16. Ans (3)



$$2T \sin \theta = w$$

$$T = \frac{w}{2 \sin \theta}$$

$$\theta \downarrow T \uparrow$$

more likely to break.

17. Ans (3)

$$F = ma$$

$$= (60)(1)$$

$$= 60 \text{ N}$$

18. Ans (2)

$$\frac{\Delta R}{R} = \pm \frac{\Delta V}{V} \pm \frac{\Delta I}{I}$$

$$= \pm \frac{8}{800} \pm \frac{3}{100} = 4\%$$

19. Ans (2)

$$\text{L.C.} = M \left[1 - \frac{8}{10} \right]$$

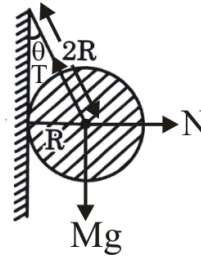
$$= \frac{1}{10} \left[\frac{2}{10} \right] \text{ cm} = 0.02 \text{ cm}$$

SECTION-B

38. Ans (4)

$$\text{Resultant intensity} = 4I \cos^2 \left(\frac{\phi}{2} \right)$$

39. Ans (3)



$$\therefore \sin \theta = \frac{R}{2R} = \frac{1}{2}$$

$$\therefore \cos \theta = \frac{\sqrt{3}}{2}$$

$$T \cos \theta = Mg \quad \dots(1)$$

$$T \sin \theta = N \quad \dots(2)$$

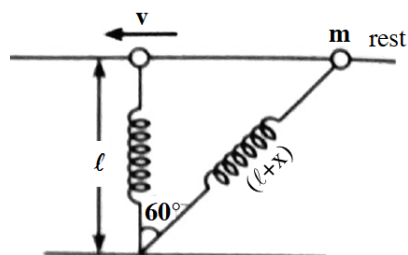
$$\text{So } (2) / (1)$$

$$\frac{N}{Mg} = \tan \theta$$

$$N = -Mg \tan 30^\circ$$

$$= \frac{Mg}{\sqrt{3}}$$

40. Ans (2)



$$\therefore (l+x) \cos 60^\circ = l$$

$$(l+x) = 2l \Rightarrow x = l$$

By COME law

$$\frac{1}{2} k (l)^2 + 0 = + \frac{1}{2} m v^2$$

41. Ans (2)

$$P = \frac{mgh}{t} \Rightarrow 2000 = \frac{m(10)(10)}{60}$$

$$m = 1200 \text{ kg}$$

Means volume of water 1200 litre.

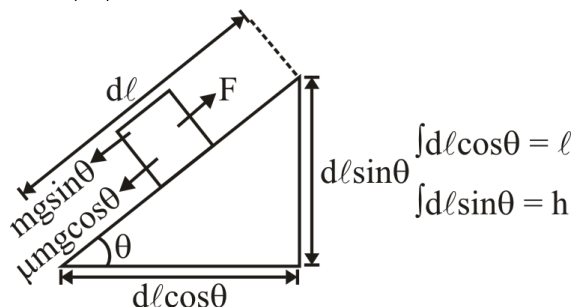
42. Ans (3)

$$PE = \frac{1}{2} kx^2 = \frac{(kx)^2}{2k} = \frac{F^2}{2k}$$

For same F

$$PE \propto \frac{1}{k}$$

43. Ans (3)



For small displacement $d\ell$ work done by F .

$$dw = (mg \sin \theta) d\ell + (\mu mg \cos \theta) d\ell$$

$$w = mg \left[\int d\ell \sin \theta + \mu \int d\ell \cos \theta \right]$$

$$= mg[h + \mu \ell]$$

45. Ans (3)

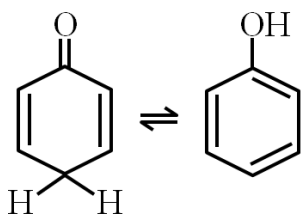
$$(\tau) = \frac{m^1}{m^1 L^{-1} T^{-1} L^1}$$

$$(\tau) = T^1$$

SUBJECT : CHEMISTRY

SECTION-A

51. Ans (4)



82. Ans (4)

NCERT Page No. 210

83. Ans (3)

NCERT Page No. 215

SECTION-B

92. Ans (3)

$$\pi \propto T$$

$$\frac{\pi_1}{\pi_2} = \frac{T_1}{T_2}$$

$$\therefore \frac{\pi_1}{2} = \frac{546 \text{ K}}{273 \text{ K}}$$

$$\therefore \pi_1 = 4 \text{ atm}$$

95. Ans (3)

TG: @Chalnaayaaar

Total volume of the buffer solution

$$= 300 \text{ cc} + 500 \text{ cc} = 800 \text{ cc}$$

$$\text{Number of milli moles of } \text{NH}_3 = 300 \times 0.3 = 90$$

$$\text{Molarity of } \text{NH}_3 \text{ (Base) in the buffer} = 90/800 \text{ M}$$

$$\text{Number of milli moles of } \text{NH}_4\text{Cl} = 500 \times 0.5 = 250.0$$

$$\text{Molarity of [salt] in the buffer} = 250/800 \text{ M}$$

Henderson's equation for basic buffer is:

$$\text{pOH} = -\log K_b + \log \frac{[\text{Salt}]}{[\text{Base}]}$$

$$= -\log 1.8 \times 10^{-5} + \log \frac{250/800}{90/800}$$

$$= 4.74 + \log 250/90$$

$$= 4.74 + 0.44$$

$$= 5.18$$

$$\text{pH} = 14 - \text{pOH} = 14 - 5.18 = 8.82$$

96. Ans (1)

The addition of NaOH converts equivalent amount of acetic acid into sodium acetate.

Hence,

Concen. of acetic acid after the addition of

$$\text{NaOH} = \frac{30}{70} \times 0.2 \text{ M}$$

Concen. of CH_3COONa after the addition of

$$\text{NaOH} = \frac{20}{70} \times 0.2 \text{ M}$$

Hence, Using the expression

$$\text{pH} = \text{pKa} + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$= -\log (2 \times 10^{-5}) + \log \left(\frac{20}{30} \right)$$

$$= 4.699 - 0.177 = 4.522$$

98. Ans (4)

NCERT Page No. 219

SUBJECT : BIOLOGY-I
SECTION-A

101. **Ans (2)**
NCERT (XII) Pg# 169 Para: 9.1.2
102. **Ans (3)**
NCERT (XII) Pg#167 Fig:9.1(b)
104. **Ans (3)**
NCERT (XIIth) Pg. # 167, 168
111. **Ans (1)**
NCERT (XIth) Pg. # 146
113. **Ans (2)**
NCERT-Pg. # 174 last line
114. **Ans (3)**
NCERT XI Page No. # 96
115. **Ans (2)**
NCERT-XII Pg. # 181
116. **Ans (2)**
NCERT-XII Pg. # 183
117. **Ans (3)**
NCERT-XII Pg. # 187
118. **Ans (4)**
NCERT-XII Pg. # 187
119. **Ans (1)**
NCERT-XII Pg. # 183
120. **Ans (3)**
NCERT Page No. 212
121. **Ans (4)**
NCERT Page No. 208
122. **Ans (2)**
NCERT-XII Pg. # 208
123. **Ans (2)**
NCERT Page No. 176
130. **Ans (4)**
NCERT(XII) Pg#140/151(H) Para:7.9
131. **Ans (4)**
NCERT XII, Pg. # 133, Fig.-7.6

SECTION-B

142. **Ans (3)**
NCERT-XII Pg. # 188
143. **Ans (4)**
NCERT Page No. 208
144. **Ans (4)**
NCERT-XII Pg. # 210

SUBJECT : BIOLOGY-II
SECTION-A

152. **Ans (2)**
NCERT (XIth) Pg. # 275
161. **Ans (2)**
NCERT Pg. # 274
163. **Ans (3)**
NCERT-XIth Pg# 274/275
167. **Ans (3)**
NCERT–XI, Para-1,2 Pg. # 283,284
168. **Ans (4)**
NCERT (XI) Pg. # 280; para-2
169. **Ans (1)**
NCERT (XI) Pg. # 281; para-2
175. **Ans (3)**
NCERT (XI) Page No. # 284
182. **Ans (2)**
NCERT Pg. # 138
185. **Ans (4)**
NCERT - 132

SECTION-B

191. **Ans (4)**
NCERT-XI, PG No. 274
192. **Ans (2)**
NCERT XI Pg. # 286
193. **Ans (4)**
NCERT (XIth) Pg. # 286
194. **Ans (3)**
NCERT (XIth) Pg. # 284